

Dimitrios Papageorgiou

Associate Professor of Nonlinear and Fault-Tolerant Control

Technical University of Denmark (DTU)
Department of Electrical and Photonics Engineering
Control, Robotics and Embodied AI (CREA)
Building 326, Room 124 · 2800 Kgs. Lyngby, Denmark
dimpa@dtu.dk

ORCID: 0000-0002-4900-4083
H-index: 15 Google Scholar · 14 Scopus
Citations: 520 Google Scholar · 498 Scopus

ACADEMIC POSITIONS

2023 - Present **Associate Professor of Nonlinear and Fault-Tolerant Control**
Technical University of Denmark (DTU), Department of Electrical and Photonics Engineering

2020 - 2023 **Assistant Professor in Automation and Control**
Technical University of Denmark (DTU), Department of Electrical and Photonics Engineering

2017 - 2020 **Postdoctoral Researcher**
Technical University of Denmark (DTU), Department of Electrical Engineering

EDUCATION

2017 **PhD**
Technical University of Denmark (DTU)
Thesis: High Accuracy Nonlinear Control and Estimation for Machine Tool Systems

2014 **MSc in Electrical Engineering**
Technical University of Denmark (DTU)
Automation and Robot Technology
GPA: 11.1/12

2011 **Diploma in Electrical and Computer Engineering**
National Technical University of Athens
Information Technology
Five-year integrated degree
GPA: 7.08/10

RESEARCH INTERESTS

Nonlinear control theory · Sliding mode control · Fault diagnosis and fault-tolerant control · Safe estimation · Condition monitoring · Maritime autonomy · Relieable and resilient operation

RESEARCH PROJECTS & FUNDING

2026–2029 **TRANSITION** · Participant
Resilient Renewable Power Generation: Hybrid Power Plants with Distributed Control
Independent Research Fund Denmark · Green Research

2026–2029 **Smart AI** · Principal Investigator
Learning-based parameter estimation in high-dimensional spaces for marine engine commissioning
Innovation Fund Denmark · Industrial PhD

2026–2029 **SAFE-LIFT** · Principal Investigator
Learning-based dimension lifting for singularity avoidance in dynamical systems
Danish Data Science Academy · DDSA PhD Scholarship

2026–2029 **SAFE-T** · Principal Investigator
Singularity-Aware Framework for Estimation via Topology Shaping
Independent Research Fund Denmark · Research Project 2

2025–2028 **U-Shield** · Task Leader
Underwater Safeguarding High-value Infrastructure Enforcing Lightweight Drones
Innovation Fund Denmark · Grand Solutions

2024–2025 **ENDYPOS** · Principal Investigator
Enhanced dynamic positioning for in-harbour navigation via Modelling and Monitoring
Orients Fond

2024–2028 **EvoRoads** · Task Leader
Evolutionary Solutions for Realising a Holistic Safe System Approach for All Road Users
European Union · Horizon Europe Research and Innovation

2024–2027 **SLGreen** · Task Leader
Green Transition of the Blue Denmark through Digitalization, Decarbonization, and Safety
Innovation Fund Denmark · Grand Solutions

2023–2024 **COLECT** · WP Leader
Control and Learning of Contact Transitions
Odense Robotics

2023–2025 **AI-Navigator** · Participant
Advanced Navigation Assistance System for Highly Automated Vessels
Den Danske Maritime Fond

2022–2027 **HyRel** · WP Leader
Decreased Cost of Energy (CoE) from wind turbines by reducing pitch system faults
Energy Technology Development and Demonstration Programme

2019–2022 **ShippingLab** · Task Leader
ShippingLab Autonomy
Innovation Fund Denmark, Den Danske Maritime Fond, Orients Fond · Grand Solutions

AWARDS & DISTINCTIONS

2023 Best teacher of the semester, MSc course 34746 – Robust and Fault-Tolerant Control

2018 Best teacher of the semester, MSc course 31320 – Robust and Fault-Tolerant Control

PROFESSIONAL ACTIVITIES & MEMBERSHIPS

2026 - Present **Vice Chair**
IEEE Control Systems Society, Technical Committee on Variable Structure and Sliding Mode Control

2024 - Present **Member**
IEEE Control Systems Society

TEACHING

2018 - Present **Robust and Fault-Tolerant Control**
MSc course, 10 ECTS

2018 - Present **Topics in Advanced Control**
PhD course, 5 ECTS

2020 - Present **Linear Control Design 1**
BSc course, 5 ECTS

2023 **Advanced Fault-Tolerant and Switching Control**
PhD course, 7.5 ECTS

2020 **Advanced Topics in Diagnosis and Fault-Tolerant Control**
PhD course, 7.5 ECTS

2020 - 2021 **Statistical Methods in Change Detection and Machine Learning**
PhD course, 5 ECTS

2021 **Reinforcement Learning for Planning and Decision-Making**
PhD special course, 7.5 ECTS

INVITED TALKS & VISIBILITY

2026 **Bridging Virtual and Physical Reality: Ensuring Valid and Relevant Maritime Data**
Data, Algorithms and the Physical Maritime Reality, IDA, Copenhagen

2024 **Sailing in the Time of Artificial Intelligence – Examining Autonomous Solutions**
Panel discussion, 5th Monaco Hydrogen Round Table, Monaco Energy Boat Challenge

2024 **Advances on Industrial Automation and Control – Math, AI and the Holy Grail of Auto-Commissioning**
DAU Conference, Copenhagen

2023 **Soft Robot Control Demonstration**
DTU Electro delegation leader, European Robotics Forum (ERF)

2023 **Neuromorphic Control for Robotic Manipulation**
Denmark's Rotary Club visit to DTU Electro AUT

2023 **Greenhopper: Denmark's First Autonomous Ferry**
Business France Denmark visit to DTU Center for Collaborative Autonomous Systems (CCAS)

2023 **Neuromorphic Control for Robotic Manipulation**
Business France Denmark visit to DTU Center for Collaborative Autonomous Systems (CCAS)

2022 **Machine Learning Enhancement of Digital Twin for Ship Maneuvering Prediction**
DAU Conference, Copenhagen

2021 **Super Twisting Sliding Mode Control in Periodic Systems – Stability and Tuning**
SMC Online Seminar

2019 **High Accuracy Nonlinear Control and Estimation for Machine Tool Systems**
IDA Conference, Aarhus

2018 **Maskiner kompenserer selv for slitage**
IDA Conference, Copenhagen

SUPERVISION

PhD supervision: 9 candidates · 6 current · 3 completed
Postdoctoral supervision: 1 researchers · 0 current · 1 former
Student supervision: 75 completed projects · 55 MSc theses · 9 BSc projects

ACADEMIC SERVICE

• Reviewer: Automatica, IEEE Transactions on Automatic Control, Control Engineering Practice, IEEE Transactions on Control Systems Technology, IEEE Control Systems Letters, International Journal of Control, IEEE Transactions on Industrial Electronics, Mechatronics, and International Journal of Advanced Robotic Systems.
• Editorial service: Guest Editor, Control Engineering Practice.